

RESEARCH PORTFOLIO

Multimodal Human Behaviour Analysis

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Research Statement

Artificial intelligence initially attracted me because I wanted to understand how intelligent systems learn. During my MSc in Artificial Intelligence, I gradually realised that many of the principles underlying AI originated from attempts to understand the human brain. I became increasingly interested in using computational methods to better understand cognition, behaviour, and mental processes.

My current research focuses on multimodal artificial intelligence for analysing human behaviour. By combining computer vision, behavioural modelling, and temporal analysis, I aim to investigate how multiple behavioural signals can provide insights into emotional dynamics, stress responses, and behavioural adaptation during human interactions.

My long-term research goal is to contribute to computational neuroscience and brain-inspired artificial intelligence by developing computational models that improve our understanding of cognition while inspiring more biologically plausible intelligent systems.

Research Project

Multimodal Human Behaviour Analysis

Abstract

Understanding human behaviour requires integrating multiple sources of information rather than analysing isolated behavioural signals. Human posture, facial expressions and temporal dynamics each capture different aspects of cognitive and emotional processes. This project investigates how multimodal artificial intelligence can combine these complementary signals to analyse behavioural adaptation, emotional dynamics, stress responses and threat perception during human interactions.

The proposed framework integrates pose estimation, facial expression analysis and temporal modelling within a unified computational pipeline. Rather than focusing solely on prediction, the project aims to explore richer behavioural representations capable of supporting future research in human behaviour analysis, mental health and computational neuroscience.

This work is currently under development.

Motivation

Human behaviour is complex and cannot be fully understood through a single source of information. Cognitive and emotional states are expressed simultaneously through body posture, facial expressions, movement and temporal behavioural patterns. Analysing these modalities together may reveal behavioural mechanisms that remain hidden when each signal is studied independently.

Recent advances in artificial intelligence provide new opportunities to integrate these heterogeneous sources of information into unified computational models. My motivation is to explore how multimodal AI can contribute not only to more robust behavioural analysis but also to a deeper understanding of human cognition.

Methodology

Overall Framework

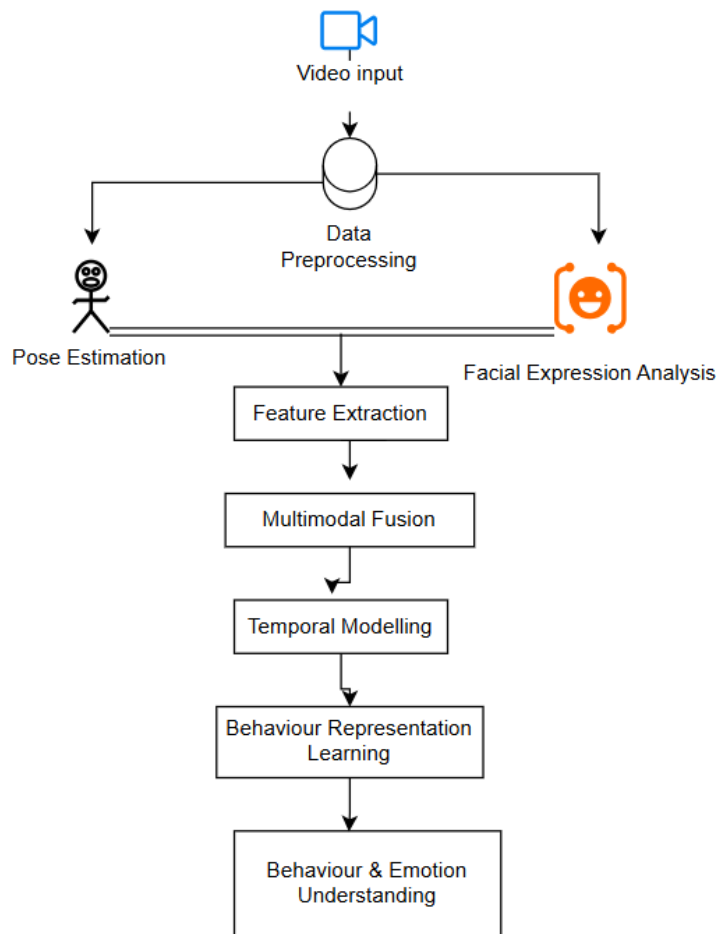
The proposed framework aims to analyse human behaviour by integrating multiple complementary behavioural modalities within a unified multimodal artificial intelligence pipeline. Rather than relying on a single source of information, the framework combines body posture, facial expressions, and temporal dynamics to build a richer representation of human behaviour.

The overall methodology is composed of five main stages. First, video sequences are acquired and preprocessed to ensure consistent input quality. Behavioural information is then extracted independently from different modalities. Pose estimation captures body movements and posture, while facial analysis extracts facial landmarks and expression-related features. These complementary behavioural signals are subsequently integrated through a multimodal fusion module to generate a unified behavioural representation.

Because human behaviour evolves over time, temporal modelling is incorporated to capture behavioural dynamics across consecutive frames rather than analysing isolated images. The resulting representation is then used to investigate behavioural adaptation, emotional dynamics, stress responses, and threat perception during human interactions.

The framework is designed to remain modular, allowing different models and feature extraction methods to be incorporated as the project evolves. This flexibility facilitates future improvements and enables the integration of additional behavioural modalities when required.

Overall Pipeline



Why this methodology?

The proposed methodology is motivated by the observation that human behaviour cannot be fully characterised using a single behavioural cue. Facial expressions, body posture, and movement dynamics each capture different aspects of human cognition and emotion. Integrating these modalities enables a more comprehensive understanding of behavioural responses while reducing the limitations associated with analysing individual signals independently.

Furthermore, incorporating temporal modelling allows the framework to capture behavioural evolution over time, providing additional information that static image analysis cannot represent. This makes the framework particularly suitable for analysing complex human interactions where behaviour continuously changes throughout an event.

Pose Estimation

Pose estimation is used to capture the body's spatial configuration during human interactions by detecting anatomical keypoints such as the head, shoulders, elbows, wrists, hips, knees, and ankles. These landmarks provide a structured representation of body posture and movement that is independent of appearance, enabling the analysis of gestures, body orientation, and behavioural responses. The extracted pose features serve as one of the primary behavioural modalities within the proposed multimodal framework.

Facial Expression Analysis

Facial expression analysis extracts visual features that reflect emotional and behavioural states during interactions. The framework identifies facial landmarks and expression-related characteristics that provide information about affective responses and non-verbal communication. These features complement body posture by capturing behavioural cues that may not be observable through movement alone, thereby enriching the overall behavioural representation.

Temporal Modelling

Human behaviour is inherently dynamic and cannot be fully understood from isolated frames. Temporal modelling is therefore incorporated to capture behavioural evolution over time by analysing sequences of extracted behavioural features. This component aims to identify temporal patterns associated with behavioural adaptation, emotional changes, and responses to different interaction contexts, providing a more comprehensive representation than static analysis alone.

Multimodal Fusion

Each behavioural modality captures different aspects of human behaviour. The multimodal fusion module combines pose, facial, and temporal representations into a unified feature space, allowing complementary information from each modality to be jointly exploited. By integrating these heterogeneous signals, the proposed framework aims to generate richer behavioural representations that can support the analysis of complex human interactions.

Behaviour Representation

The final stage of the framework produces a unified behavioural representation by integrating information from all modalities. Rather than focusing solely on classification or prediction, this representation is intended to capture meaningful behavioural patterns that can support the study of emotional dynamics, stress responses, behavioural adaptation, and future applications in human behaviour analysis, mental health, and computational neuroscience.

Data Source

To evaluate the proposed framework, the current implementation uses publicly available boxing videos as a proof-of-concept. These videos provide dynamic human interactions involving body movements, facial expressions, and rapid behavioural changes, making them suitable for investigating multimodal behavioural analysis.

At this stage, the videos are used to validate the overall processing pipeline, including pose estimation, facial analysis, temporal modelling, and multimodal feature integration. As the project progresses, the framework will be evaluated using more diverse interaction scenarios and, where appropriate, publicly available behavioural datasets.

Experimental Scenario

The current implementation uses boxing videos as an experimental scenario to evaluate the proposed multimodal framework. Boxing provides a particularly suitable environment for behavioural analysis because it involves continuous human interaction under conditions of heightened stress, threat perception, rapid decision-making, and behavioural adaptation.

During a fight, athletes constantly adjust their posture, facial expressions, movements, and defensive strategies in response to their opponent's actions. These behavioural changes occur dynamically over time and reflect cognitive processes such as attention, anticipation, perception, and reaction. Consequently, boxing offers a rich source of multimodal behavioural information that can be analysed through body posture, facial expressions, and temporal dynamics.

Using boxing videos as an initial proof-of-concept allows the proposed framework to be evaluated in a challenging environment before extending it to broader applications involving everyday social interactions, mental health, and computational neuroscience.

Software Environment

The framework is implemented in Python using widely adopted computer vision and deep learning libraries.

The current implementation includes:

- Python for data processing and model development.
- TensorFlow for deep learning models.
- OpenCV for image and video processing.
- MediaPipe for pose estimation and facial landmark extraction.
- NumPy and Pandas for data manipulation.
- Matplotlib for visualisation and evaluation.

The modular implementation allows additional models and libraries to be incorporated as the project evolves.

Data Preprocessing

Before behavioural analysis, all videos undergo several preprocessing steps to ensure data consistency.

These include:

- Video frame extraction.
- Image resizing and normalization.
- Landmark extraction.
- Temporal sequence generation.

- Feature standardisation.

The resulting data are organised into multimodal feature sequences suitable for temporal modelling.

Model Development

The current framework consists of multiple independent modules that process different behavioural modalities before integrating them into a unified representation.

The implementation currently includes:

- Pose estimation.
- Facial feature extraction.
- Feature engineering.
- Temporal modelling.
- Multimodal feature fusion.

Each component has been developed independently to facilitate future experimentation and model optimisation.

Current Progress

Since the project is ongoing, this chapter summarises the current development status and preliminary observations.

Completed Components

The following milestones have been completed:

- ✓ Literature review
- ✓ Overall system architecture
- ✓ Video preprocessing pipeline
- ✓ Pose estimation module
- ✓ Facial expression analysis module
- ✓ Feature extraction pipeline

These components establish the foundation for multimodal behavioural analysis.

Current Development

The current work focuses on integrating individual behavioural modalities into a unified computational framework.

The ongoing development includes:

- Multimodal feature fusion.
- Temporal sequence modelling.

- Behaviour representation learning.
- Experimental evaluation.
- Model optimisation.

Future iterations will further improve robustness and generalisation across different interaction scenarios.

Preliminary Results

The current prototype successfully integrates person detection, pose estimation, face detection, and facial expression analysis within a unified multimodal pipeline. The system identifies multiple participants, assigns a unique identity to each boxer, estimates upper-body landmarks (including the shoulders, elbows, and wrists), and detects visible faces throughout the interaction.

Beyond visualising these outputs directly on the video stream, the framework automatically generates a time-stamped behavioural log recording the facial expression predicted for each participant at different moments during the interaction. This temporal output provides a structured representation of behavioural changes over time rather than limiting the analysis to frame-by-frame visualisation.

An example of the generated output is shown below:

00:00 Boxer 1 → Fear

00:03 Boxer 1 → Fear

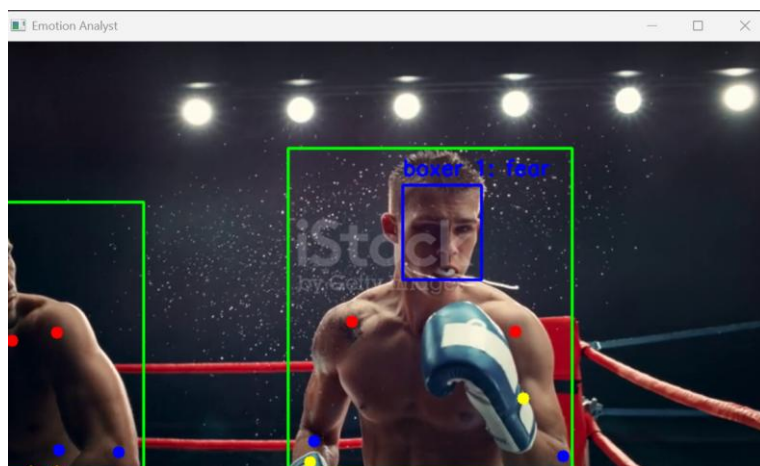
00:06 Boxer 1 → Sad | Boxer 2 → Fear

00:11 Boxer 1 → Sad | Boxer 2 → Sad

00:13 Boxer 1 → Neutral

00:15 Boxer 1 → Happy

These preliminary results demonstrate that the proposed framework can simultaneously detect multiple participants, estimate body posture, analyse facial expressions, and produce temporally organised behavioural information.



Discussion

The current emotion labels originate from a pre-trained facial expression recognition model and therefore represent predicted facial-expression categories rather than verified emotional states. The objective of this component is not to infer a participant's true emotions, but to provide one behavioural modality that will later be integrated with body posture and temporal dynamics.

Future work will therefore focus on multimodal fusion and temporal modelling to obtain richer and more robust behavioural representations.

Conclusion

This technical report presented an ongoing research project investigating the use of multimodal artificial intelligence for human behaviour analysis. By integrating pose estimation, facial expression analysis, temporal modelling, and multimodal fusion, the proposed framework aims to construct richer behavioural representations than can be obtained from individual modalities alone.

The preliminary implementation demonstrates the feasibility of combining multiple behavioural signals within a unified computational framework while highlighting the challenges associated with modelling complex human interactions. Although the project is still under development, it establishes a solid foundation for exploring how multimodal AI can contribute to a deeper understanding of behaviour beyond simple prediction or classification.

Future Work

The next stage of this research will focus on improving both the robustness of the framework and the quality of the learned behavioural representations. Future developments include:

- Improving multimodal feature fusion and representation learning.
- Investigating Transformer-based architectures for temporal modelling.
- Exploring graph neural networks to capture relationships between behavioural features.
- Integrating self-supervised learning techniques to reduce dependence on labelled data.
- Extending the framework to additional behavioural modalities.
- Evaluating the framework on more diverse human interaction scenarios.
- Investigating applications in human behaviour analysis, cognition, and mental health.

Research Perspective

This project represents the first step toward a broader research direction that I hope to pursue in the future. While the current work focuses on multimodal behaviour analysis, my long-term objective is to investigate how artificial intelligence can help us better understand the mechanisms underlying human cognition and behaviour.

I believe that human behaviour emerges from the interaction of multiple cognitive processes, including perception, attention, memory, emotion, and decision-making. Advances in artificial

intelligence provide new opportunities to model these complex interactions and uncover behavioural patterns that are difficult to observe directly. At the same time, a deeper understanding of these mechanisms can inspire the development of more biologically plausible and human-centred AI systems.

For me, the relationship between artificial intelligence and the study of cognition is reciprocal: insights from cognitive science and neuroscience can guide the development of more brain-inspired AI, while advances in AI can provide powerful computational tools for exploring cognition, behaviour, and mental health. I hope to contribute to this interdisciplinary research direction by developing computational models that not only analyse behaviour but also support scientific discovery.

Code Availability

The implementation of the proposed framework is publicly available and actively maintained on GitHub. The repository contains the source code for the multimodal behaviour analysis pipeline, including pose estimation, facial expression analysis, temporal modelling, data preprocessing, and visualisation modules.

<https://github.com/narcisse2/Multimodal-Human-Behaviour-Emotion-Analysis-on-going->

Related Repositories

In addition to this project, I have developed several machine learning and artificial intelligence projects during my MSc, including:

Project	Description
Autoencoder & Representation Learning	Deep learning models for feature extraction and anomaly detection
Siamese Neural Networks	Similarity learning and image comparison
CNN Image Recognition	Computer vision with transfer learning
NLP Sentiment Analysis	Word2Vec and neural network models
Human Behaviour Analysis	Multimodal AI framework (current project)

GitHub Profile :

<https://github.com/narcisse2>

Kaggle:

If applicable, supplementary notebooks and experiments are available on Kaggle.

<https://www.kaggle.com/machinelearningttt>

Research Resources

This project is continuously updated as new modules are implemented. The online repositories provide:

- Source code
- Experimental notebooks
- Documentation
- Model architectures
- Future developments